RESEARCH PROBLEM

Pre-Harvest Crop Yield and Total Production Estimation

Initial case study: Pomegranate (Punica granatum L.)

1. Manual assessment

Fruit counting and field assessment are time-consuming and may not provide a representative picture of the complete crop.

2. Incomplete visibility

Leaves, branches and overlapping fruits can hide part of the crop, making visible fruit observations incomplete.

3. Fragmented AI evidence

Existing selected studies address growth stage, detection, monitoring or fruit size, but do not connect these outputs to orchard-level production.

Problem Statement

Farmers need a reliable way to estimate the expected crop yield and total production of a standing pomegranate crop before harvest. Manual counting and assessment of fruits are labor-intensive and may miss fruits that are partially or completely hidden by leaves and branches. The selected literature demonstrates useful capabilities such as growth-stage recognition, fruit detection, on-tree monitoring, and physical fruit-size estimation, but these capabilities are not integrated into a single validated pipeline that converts field observations into a dependable tree-level or orchard-level production estimate.

Research need

Develop an AI-based approach that can analyze field images or video, estimate visible fruit load, address incomplete visibility, aggregate information across trees or rows, and produce a pre-harvest estimate of expected total production.

Scope Boundary

Scope	Pre-harvest yield + total production estimation	**Excluded**	Price forecasting, lease valuation, contract pricing, supply-chain optimization

Framed as an integration gap within the selected seven-paper literature set; not a claim that yield-estimation research does not exist elsewhere.